

व्यावसायिक परीक्षण रिपोर्ट (प्रारंभिक) संख्या / No. : ICE/NRFMTTI, HISAR/2025-26/
1/1/3306
COMMERCIAL TEST REPORT (INITIAL) माह / Month : April, 2025

(यह परीक्षण रिपोर्ट 30/04/2032 तक वैध है। / THIS TEST REPORT IS VALID UP TO: 30/04/2032)



**FARMWORLD, FWMT420DEBR,
POWER WEEDER**



भारत सरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि एवं किसान कल्याण विभाग

Department of Agriculture and Farmers Welfare

उत्तरी क्षेत्र कृषि मशीनरी प्रशिक्षण एवं परीक्षण संस्थान

Northern Region Farm Machinery Training and Testing Institute

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[ISO 9001:2015 CERTIFIED]

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ICE/NRFMTTI, HISAR/ 2025-26/1/1/3306	FARMWORLD, FWMT420DEBR, POWER WEEDER (COMMERCIAL-INITIAL)
	[THIS REPORT IS VALID UP TO: 30/04/2032]

Type : Self Propelled, Walk Behind Type
Make : FARMWORLD
Model : FWMT420DEBR
Name and address of manufacturer (apa) : Farmworld Agrotech Private Limited
Plot No. 373, Road-U, Khirasara Industrial Estate
Taluka: Lodhika, Rajkot-360021 (Gujarat)
Name and address of applicant (apa) : Farmworld Agrotech Private Limited
Plot No. 373, Road-U, Khirasara Industrial Estate
Taluka: Lodhika, Rajkot-360021 (Gujarat)
Test Conducted at : Government of India,
Northern Region Farm Machinery
Training and Testing Institute,
Tractor Nagar, Sirsa Road,
Hisar-125 001 (Haryana)

THIS TEST REPORT VALID UP TO : 30th April, 2032

[vide F. No. 13-22/2020- M&T (I&P) dated 12.12.2023]

Report No. ICE/NRFMTTI, HISAR/2025-26/1/1/3306

Month : April

Year : 2025



GOVERNMENT OF INDIA
NORTHERN REGION FARM MACHINERY TRAINING & TESTING INSTITUTE
TRACTOR NAGAR, SIRSA ROAD, HISAR-125001 (HARYANA)
[ISO 9001:2015 CERTIFIED]

Type of Test : COMMERCIAL - INITIAL

Period of Test : February, 2025 to March, 2025

Test Report No. : ICE/NRFMTTI, HISAR/2025-26/1/1/3306

Month and Year : April, 2025

-
- i) The results reported in this report are observed values and no corrections have been applied for atmospheric and site conditions.
- ii) The data given in this report pertains to the particular machine submitted by the applicant for test.
- iii) The results presented in this report do not in any way attribute to the durability of the machine.
- iv) **This report should not be reproduced in part or full without prior permission of the Director, Northern Region Farm Machinery Training & Testing Institute, Hisar-125001.**
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SELECTED CONVERSIONS

1. **Force**
 - 1 kgf = 9.80665 N
 - = 2.20462 lbf
2. **Power**
 - 1 hp = 1.01387 Metric hp (Ps)
 - = 745.7 W
 - 1 Ps = 735.5 W
3. **Pressure**
 - 1 psi = 6.895 kPa
 - 1 kgf/sq.cm = 98.067 kPa = 735.56 mm of Hg
 - 1 bar = 100 kPa = 10 N/sq.cm.
 - 1 mm of Hg = 1.3333 m-bar
4. **Area**
 - 1 ha = 10000 m² = 2.471 Acres



ICE/NRFMTTI, HISAR/ 2025-26/1/1/3306	FARMWORLD, FWMT420DEBR, POWER WEEDER (COMMERCIAL-INITIAL)
	[THIS REPORT IS VALID UP TO: 30/04/2032]

IMPORTANT

This machine named FARMWORLD, FWMT420DEBR, power weeder was submitted at this institute “to test only with the tilling and earthing up device attached therewith in the dry land operation, that too, in the broad day light with sufficient visibility”.



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1. SCOPE OF TEST

The scope of the test was limited to check and assess the following :-

- 1.1 Specifications and other data furnished by the applicant.
- 1.2 Engine performance test
- 1.3 Mechanical vibration measurement
- 1.4 Noise level measurement
- 1.5 Air cleaner oil pull-over test
- 1.6 Field performance with rotary tiller
- 1.7 Operator's comfort and safety
- 1.8 Ease of operation and handling
- 1.9 Breakdowns and repairs
- 1.10 Wear of critical components
- 1.11 Hardness & chemical analysis of rotor blades

2. METHOD OF SELECTION

- i) Test requested by : The applicant
- ii) Selected for test by : NRFMT&TI, Hisar
- iii) Method of selection : Randomly selected by the representative of the institute through Virtual mode.
- iv) Serial number of the available machines at the time of Random selection of the test sample : The test sample was selected out of five number of samples, made available by the applicant for random selection.

Sr. No.	Sr. No. of machine
1.	GJ/6774/3538/2024/0470
2.	GJ/6774/3538/2024/0473
3.	GJ/6774/3538/2024/0455
4.	GJ/6774/3538/2024/0465
5.	GJ/6774/3538/2024/0458

- v) Serial number of sample selected : **GJ/6774/3538/2024/0465**

3. TEST CODE/PROCEDURE

IS:4905-2015	:	Methods for Random Sampling and Randomization Procedures
IS:15925-2012	:	Walk – Behind Powered Rotary Tillers – Definitions, Safety Requirements and Test Procedures
IS:9935-2002 (Reaffirmed 2022)	:	Power Tiller – Test code
IS:9980-1988 (Reaffirmed 2019)	:	Guidelines for Field Performance and Haulage Tests of Power Tillers
IS:12036-1995 (Reaffirmed 2019)	:	Agricultural Tractors-Test Procedures – Power Tests for Power take-off
IS:6690-1981 (Reaffirmed 2022)	:	Specification for Blades for Rotavator for Power Tillers (First revision)
IS: 13539-2018	:	Power Tillers – Recommendations on Selected Performance and Other Characteristics (Second Revision)
Critical Technical Specifications	:	Critical technical specification issued by Ministry, vide letter No. 13-9/2019 - M&T (I&P) - Part dated 26.04.2019 and F. No. 9-1/2019 M&T (I&P) dated 20.08.2019.

4. SPECIFICATIONS

4.1 General

Type of machine : Self propelled, Walk Behind Type
 Make : FARMWORLD
 Model : FWMT420DEBR
 Serial No. : GJ/6774/3538/2024/0465
 Name and address of manufacturer (apa) : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U, Khirasara Industrial Estate
 Taluka: Lodhika, Rajkot-360021 (Gujarat)

Name and address of applicant (apa) : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U, Khirasara Industrial Estate
 Taluka: Lodhika, Rajkot-360021 (Gujarat)

Year of manufacture : 2024
 Country of origin : India

4.2 Details of prime mover

Name and address of manufacturer (apa) : Farmworld Agrotech Private Limited
 Plot No. 373, Road-U, Khirasara Industrial Estate
 Taluka: Lodhika, Rajkot-360021 (Gujarat)

Type : Four stroke, air cooled, single cylinder, diesel engine

Make (apa) : FW

Model (apa) : 186FA

Sr. No. : GJ/6774/3538/2024/0465

Country of origin (apa) : India

Year of manufacture (apa) : 2024

Engine speed (recommended setting), rpm (apa)

High idle speed : 3888±100

Low idle speed : 1300±100

Rated speed : 3600

Speed at maximum torque : 2880

Rated engine speed for field : 3600

4.3 Cylinder & cylinder head

Number : One

Disposition : Vertical

Bore/stroke, mm (apa) : 86/72

Capacity, cc (apa) : 418

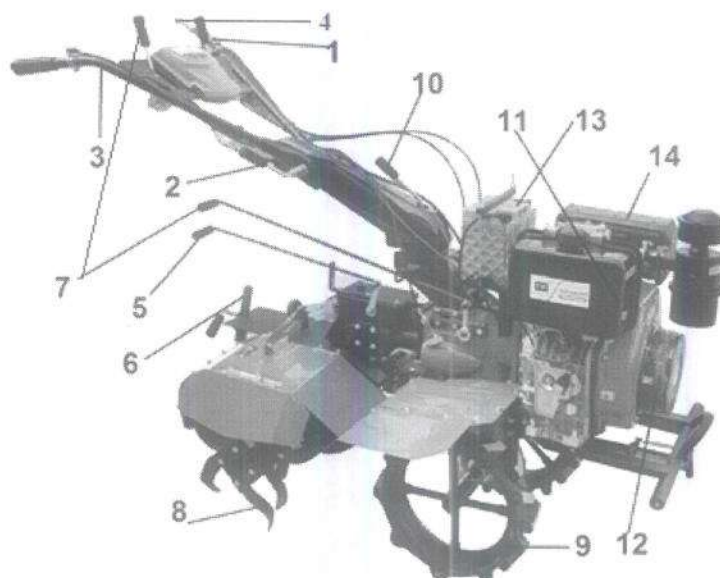
Type of valve : Over head



Valve clearance, mm (apa)	
Inlet	: 0.02 to 0.06
Exhaust	: 0.03 to 0.07
Compression ratio (apa)	: 19±1:1
4.4 Fuel supply system	
Type of fuel feed	: Gravity feed
4.4.1 Fuel tank	
Material	: Sheet metal
Capacity of fuel tank, l	: 5.0
Location of fuel tank	: On RHS of engine
Provision for draining of sediments/ water	: None
Fuel filter	: Provided in fuel tank
Fuel on/off knob	: Provided on tank
4.4.2 Governor	
Make	: Not specified
Model	: Not specified
Type	: Mechanical, centrifugal, variable speed
4.4.3 Fuel injection pump	
Make	: Not specified
Model (apa)	: LA186FA
Type	: Plunger
Sr. no.	: Not available
Method of drive	: Through camshaft
4.4.4 Fuel injector	
Make	: Not specified
Model (apa)	: LA186FA Nozzle
Type	: Multi hole
Serial no./Part no.	: Not available
Injection pressure, kg/cm ² (apa)	: 224 to 234
4.5 Air intake system	
4.5.1 Pre-cleaner	: Not provided
4.5.2 Air cleaner	
Type	: Wet, oil bath type
Make and model (apa)	: Not specified & 186 FA
Location	: In front of engine
Recommended service schedule (apa)	: After every 50 hours of operation
Recommended grade of oil (apa)	: SAE 10W30
4.6 Exhaust	
Type of silencer	: Horizontal draft, rectangular



Location of silencer : At LHS of the engine
Spark arresting device, if any : Not provided



Key Words:

- | | |
|-------------------------------|-----------------------------------|
| 1. Reverse gear lever | 2. Handle height adjusting lever |
| 3. Handle bar | 4. Main clutch engaging lever |
| 5. Rotary gear shifting lever | 6. Depth control wheel |
| 7. Main gear shifting lever | 8. Rotor blade |
| 9. Wheel | 10. Handle angle adjustment lever |
| 11. Fuel tank | 12. Recoil starter |
| 13. Battery | 14. Exhaust |

FIG. 1: FARMWORLD, FWMT420DEBR, POWER WEEDER

4.7 Lubrication system

Type	: Splash
Oil filter	: Provided
Oil capacity, l (apa)	: 1.65
Recommended grade of lubricant oil (apa)	: SAE 10W30
Oil change period (apa)	: First change after 20 hours of operation and subsequently after every 100 hours of operation or 3 months.

4.8 Cooling system

Type	: Air cooled
Dia. of blower, mm	: 260
No. of vanes	: 27

4.9 Starting system

Type	: Manual, recoil starting & self starting
Ignition system	: Compression ignition
Any other provision for easy starting	: Decompression lever is provided



	Aid for cold starting	: Not provided
4.9.1	Self starter	
	Make	: Not available
	Type	: Electrical
	Model	: HW-10-1
	Capacity/power, kW	: 12V, 1.0 kW
	Serial no.	: Not available
	Location	: On LHS of engine
4.9.2	Generator	
	Make	: Not available
	Model	: Not available
	Type	: Charge coil
	Method of drive	: Through crank shaft
4.9.3	Battery	
	Make	: Amaron
	Model	: Quanta
	Voltage	: 12 V/ 26 Ah
	Sr. No.	: M0Q02650Y215351
4.10	Power Transmission system	
	Type	: Engine power is transmitted to rotor through primary main gear box & secondary reduction gear box unit. The primary gear box consists of spur gear combination, whereas secondary reduction has bevel & pinion combination.
4.10.1	Clutch	
	Make	: BST
	Type of clutch	: Multi plate, friction, wet type
	Number of friction plate	: 5
	Dia. of friction plate, mm	: 118
	Thickness, mm	: 2.2
	Method of operation	: One hand lever is provided at LHS handle bar
	Location of clutch	: In between engine and transmission system
4.10.2	Primary (Main) gear box	
	Type	: Combination of sliding mesh & constant mesh gear box
	Make	: FW
	No. of speed	: 04 (03 forward, 1 reverse)
	Recommended grade of oil (apa)	: SAE 90
	Oil capacity, l (apa)	: 2.0



Oil change period (apa) : First change after 20 hours of operation and subsequently after every 50 hours of operation or 3 months.

Reduction Ratio

1st : 3.00:1
2nd : 2.79:1
3rd : 1.60:1
Reverse : 2.60:1

4.10.3 Secondary reduction gear box for transport wheel

Type : Gear type (Bevel and Pinion)

	1 st pair	2 nd pair
No. of teeth on drive gear pinion	10	10
No. of teeth driven gear	43	36
Reduction ratio	4.30:1	3.60:1

No. of engine revolution in one revolution of drive wheel

Forward
1st gear : 159.8
2nd gear : 42.7
3rd gear : 24.8
Reverse : 57.5

4.10.4 Secondary reduction gear box for rotary unit : The drive from gear box goes to its rotary shaft through three reduction units i.e. bevel gear set, spur gear set and chain & sprocket.

Make : FW
Type : Bevel gear drive & chain and sprocket
No. of teeth on pinion : 8
No. of teeth on bevel : 21
Reduction ratio : 2.63:1

	Forward	Reverse
No. of teeth on drive gear	22	20
No. of teeth on driven gear	27	29
Reduction ratio	1.23:1	1.45:1

Details of chain & sprocket

No. of teeth on drive sprocket : 9
No. of teeth on driven sprocket : 18
Reduction ratio : 2.00:1

No. of engine revolution in one revolution of rotor shaft

Forward : 11.1
Reverse : 8.7



Chain details, mm:
Length of chain : 890
No. of rollers : 56
Pitch : 15.10
Dia. of rollers : 10.10

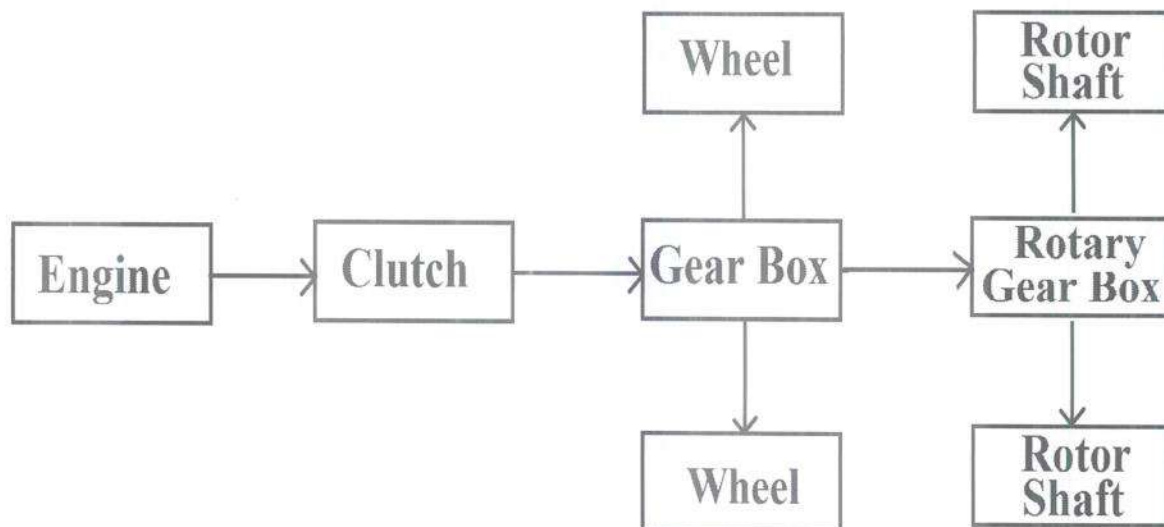


FIG. 2 : SCHEMATIC POWER TRANSMISSION DIAGRAM

4.11 Rotor

4.11.1 Rotor cover

Material (apa) : M.S. sheet

Size, mm

Length : 540 (on each side)
Width : 220 + 145 (extension on each side)
Thickness : 2.0

Method of fixing : Bolted to main frame

4.11.2 Rotor shaft

Material (apa) : SAE 1045

Type of rotor axle : Hexagonal

Width across flat of rotor axle, mm : 23

mm

Length of shaft, mm : 225 (on each side)

With tilling blade

Length of extension shaft, mm : 135 (on each side)

No. of flanges : 6 (3+3)

Type of flanges : Square

Size of flanges, mm : 95×95

Thickness of flange, mm : 4.0

Distance between two flanges, mm : 135

No. of blades on each flange : 4

No. of pocket on each shaft : --

With earthing up blade
blade
None

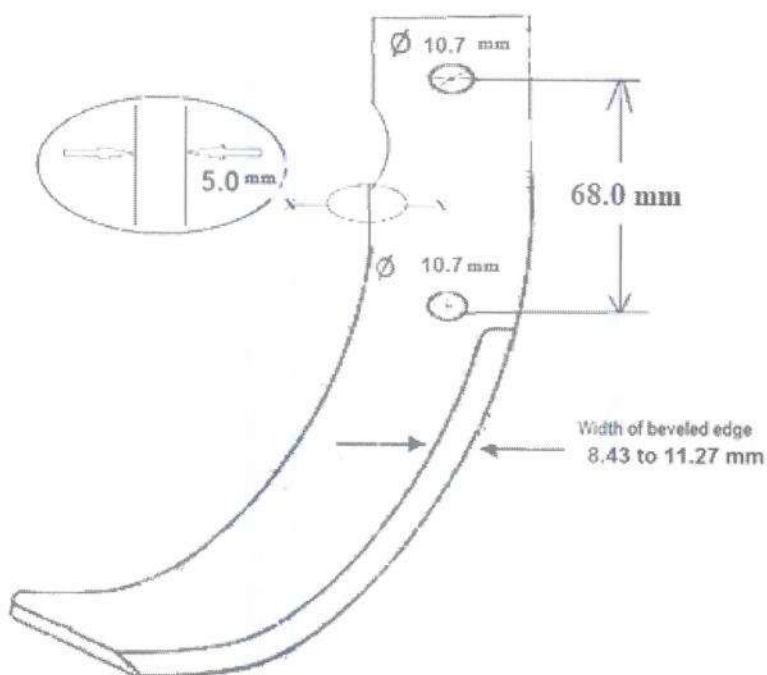


Dimensions of pocket, mm

Length	:	--	40.0
Width	:	--	9.0
Height	:	--	54.0
Thickness	:	--	2.0
No. of blade in each pocket	:	--	1
Overall length of rotor, mm	:	870	315
Diameter of rotor with blades, mm	:	350	345
Method of fixing of blade	:	By nuts & bolts	By nuts & bolts

4.11.3 Rotor blades

Number	:	24	10
Type	:	'J' shape	'C' shape
Thickness, mm	:	5.0	6.1
Width of beveled edge, mm	:	8.43 to 11.27	3.53 to 31.40
No. and dia. of hole on each blade for fixing it to the flanges, mm	:	02 and 10.70	01 and 11.02
Arrangement of blades on the axle	:	LHS & RHS mounted alternatively	Mounted in one direction in each pocket
Material of blade	:	EN42J	EN42J

**FIG. 3 : TILLING BLADE (J-Shape)**

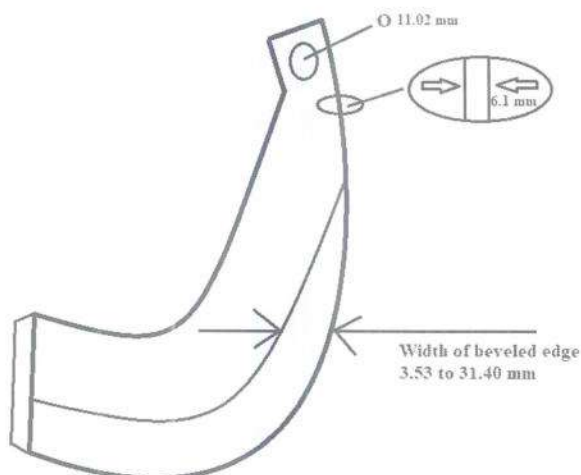


FIG. 4 : EARTHING UP BLADE (C-Shape)

- 4.12 Depth control mechanism**
- Type : Tail wheel & depth bar
- No. of wheel : 02
- No. of bar : 01
- Material (apa) : Q235A
- 4.13 Steering handle bar**
- Material (apa) : Mild steel
- Dia. of pipe, mm : 25.0
- No. of hand grips : Two
- Length of grip, mm : 115
- Material of grip (apa) : Rubber
- Dia. of grip, mm : 36.5
- Provision for handle height adjustment : Provided
- Height of handle bar from ground level, mm** : Maximum : 1350
Minimum : 860
- Provision for angle adjustment : Provided
- 4.14 Stand** : Provided
- 4.15 Transport wheel**
- Type : Hollow rim frame with solid rubber grip
- Make : Not specified
- Number : 02
- Size, mm (Dia. × Width) : 390 × 90
- Recommended pressure at 186 kg load, kPa : Not applicable
- 4.16 Controls**
- At LHS**
- i) Main clutch engaging lever
- ii) Reverse gear lever
- At Centre**
- i) Handle angle adjustment lever
- ii) Ignition on/off switch



At RHS

- i) Accelerator lever
- ii) Handle height adjustment lever
- iii) Main gear shifting lever
- iv) Rotary gear shifting lever

4.17	Overall dimensions, mm	Transport position	Working position
	Length	: 1950	1950
	Width	: 970	970 (with both types of blades)
	Height	: 1350	1350
	Mass, kg	: 186.26	
4.18	Color of machine		
	Engine	: Silver & Green	
	Main drive cover & transmission system	: Green	
	Rotary drive cover	: Green	
	Fuel tank	: Black	
	Handle bar	: Black	
4.19	Labeling /Identification plate	: Labeling plate is riveted on the machine with following information.	

FW Honour Your Trust Power Weeder	
Model	: FWMT420DEBR
Engine	: 4 Stroke Diesel
Rated power	: 6.4 kW (8.58 hp)/ 3600 RPM
Displacement	: 418
SFC	: 280 g/kWh
Tilling depth	: 100 – 250 mm
Tilling width	: 750 – 950
Sr. No.	: GJ/6774/3538/2024/0465
Engine No.	: GJ/6774/3538/2024/0465
Year	: 2024
Farmworld Agrotech Private Limited, Plot No. 373, Khirasara Industrial Estate, Rajkot, Gujarat, India www.farmworld.in	

5. FUEL & LUBRICANTS

5. FUEL & LUBRICANTS			
5.1	Fuel	:	Diesel
5.2	LUBRICANTS		
Sr. No.	Particulars	As recommended by the manufacturer	As used during the test
1.	Engine sump	SAE 10W30	Oil originally filled in the machine was not changed.
2.	Gear box	SAE 90	

ICE/NRFMTTI, HISAR/ 2025-26/1/1/3306	FARMWORLD, FWMT420DEBR, POWER WEEDER (COMMERCIAL-INITIAL)
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6. ENGINE RATING TEST AND FUEL CONSUMPTION TEST

Date of test : 28.02.2025
 Model of dynamometer : SAJ-AG-10
 Dynamometer constant : 9549.305

TABLE:1 ENGINE PERFORMANCE TEST

Brake power, kW	Engine speed, rpm	Fuel consumption			Specific energy,
		l/h	kg/h	Specific, gm/ kWh	kWh/l
a) Maximum power 2 hours test					
7.08	3701	4.09	3.44	485.88	1.73
b) Power at rated engine speed: (3600 rpm)					
6.84	3599	4.44	3.74	546.78	1.54
c) Varying load test:					
i) Torque corresponding to maximum power					
7.08	3701	4.09	3.44	485.88	1.73
ii) 85% of torque obtained at maximum power:					
6.16	3783	2.19	1.84	298.70	2.81
iii) 75% of torque defined in (ii)					
4.63	3786	1.66	1.39	300.22	2.80
iv) 50% of torque defined in (ii)					
3.08	3792	1.28	1.08	350.64	2.41
v) 25% of torque defined in (ii)					
1.54	3799	1.02	0.86	558.44	1.51
vi) Unloaded					
0.16	3802	0.79	0.66	4125.00	0.20

TABLE:2 Engine performance parameters

Parameters		Natural ambient
No load engine speed, rpm	:	3795
Crankshaft torque at max. power, Nm	:	18.30
Max. crankshaft torque, Nm	:	21.27
Engine speed at max. crankshaft torque, rpm	:	2402
Range of atmospheric conditions:		
Temperature, °C	:	25.4 to 30.8
Pressure, kPa	:	98.7 to 99.0
Relative humidity, %	:	55.7 to 73.7
Max. temperature, °C		
Engine oil	:	110.2
Fuel	:	28.5
Air intake	:	30.9
Exhaust gas	:	265
Max. smoke density	:	2.75



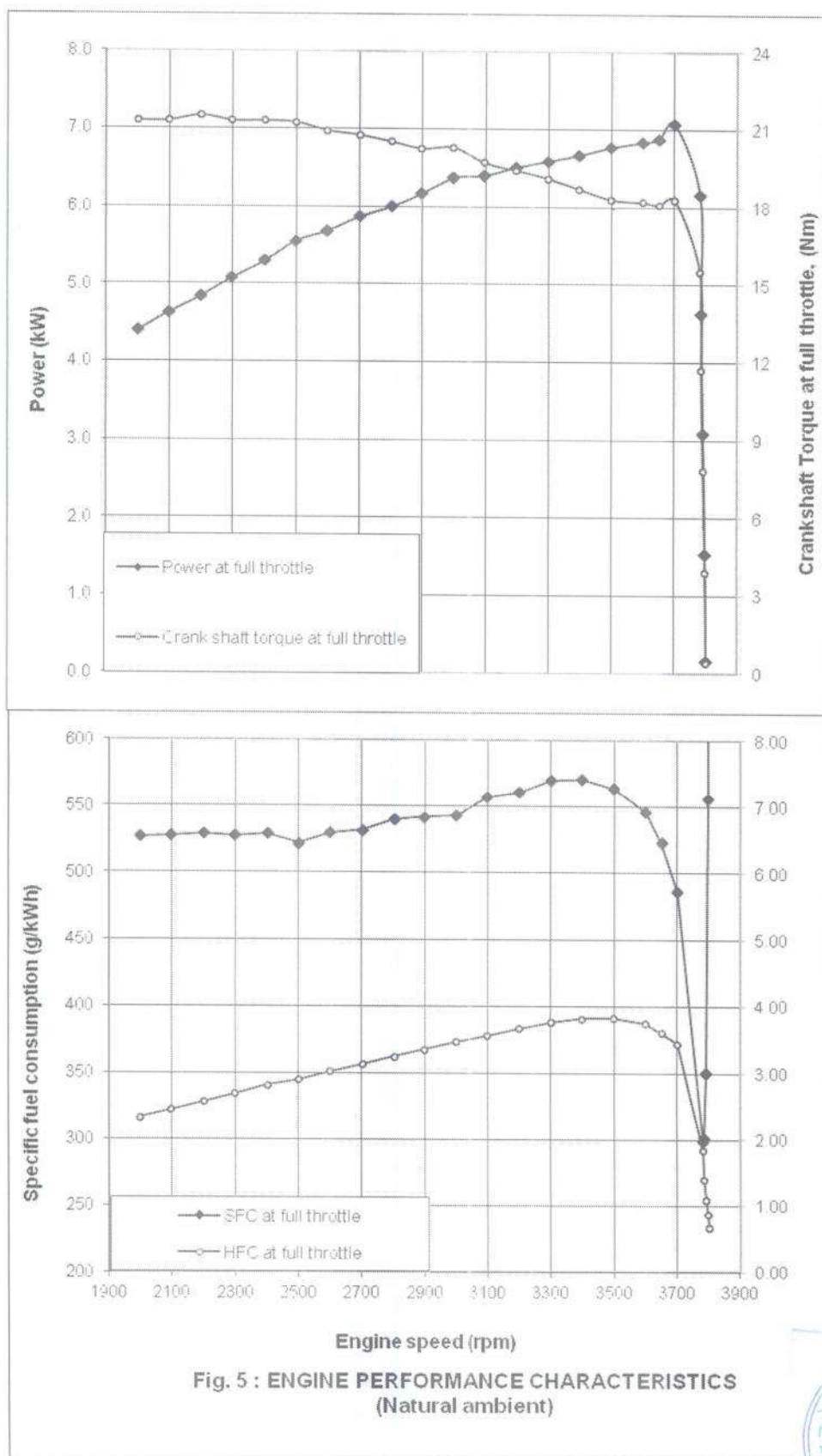
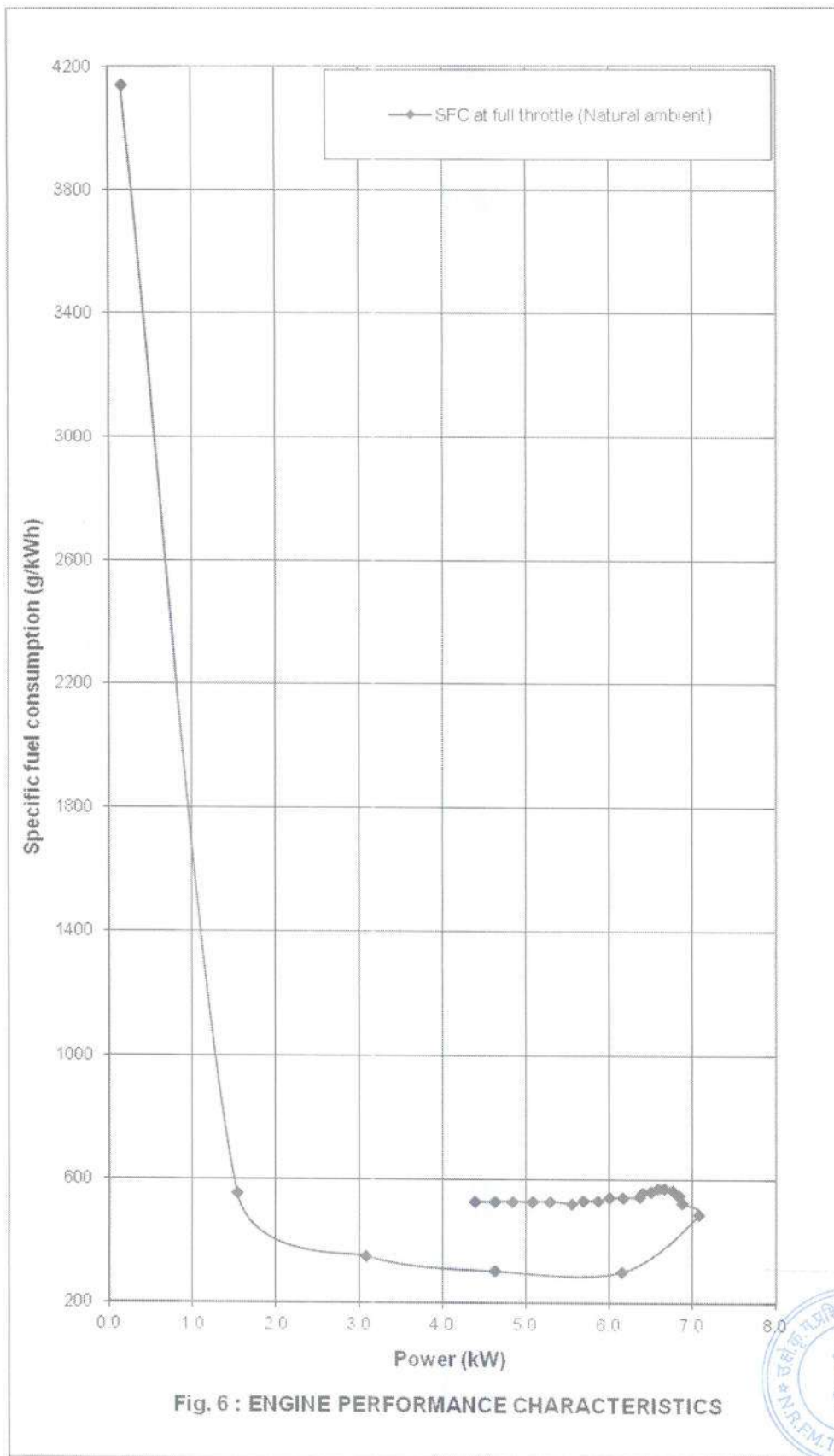


Fig. 5 : ENGINE PERFORMANCE CHARACTERISTICS
(Natural ambient)





ICE/NRFMTTI, HISAR/ 2025-26/1/1/3306	FARMWORLD, FWMT420DEBR, POWER WEEDER (COMMERCIAL-INITIAL)
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7. MECHANICAL VIBRATION TEST

The amplitude of mechanical vibration on various assemblies/components of rotary power weeder was recorded by running the machine under stationary condition with accelerator lever recommended for rated engine speed with rotary shaft in running condition. The amplitude of vibration was measured in horizontal and vertical position of the accelerometer and the results are represented in Table -3.

Table 3: Vibration Measurement

Sr. No.	Location	Vibration (microns)	
		HD	VD
1.	Handle		
	Left	410*	630*
	Right	430*	520*
2.	Air cleaner	310*	370*
3.	Clutch lever	280*	330*
4.	Gear shifting lever	440*	220*
5.	Wheel fender		
	Left	150*	180*
	Right	200*	130*
6.	Handle up/down adjustment lever	230*	250*
7.	Depth control lever	150*	140*
8.	Rotary gear lever	230*	200*
9.	Accelerator lever	220*	330*
10.	Fuel tank	170*	130*
11.	Reverse gear lever	230*	270*
12.	Rotary gear lever	230*	200*
13.	Starter switch	310*	220*
14.	Rotary cover		
	Left	210*	170*
	Right	180*	220*
15.	Handle left/right lever	180*	120*

Remarks: The amplitude of mechanical vibration on the various assemblies and sub assemblies as marked (*) are considered to be on higher side.

8. NOISE LEVEL MEASUREMENT

8.1 Noise at By-Standers position

Date of test	: 19.02.2025
Type of track	: Concrete
Background noise level, dB (A)	: 52
Atmospheric conditions	
Temperature, °C	: 33.5
Pressure, kPa	: 98.8
Relative humidity, %	: 28.4
Wind velocity, m/s	: 1.0 to 1.4



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TEST DATA:

Sr. No.	Gear Used	Travelling speed before acceleration (kmph)	Noise level, dB (A)	
			Silencer facing microphone	Silencer facing away from microphone
1.	G-1	1.27	80	80
2.	G-2	4.74	80	80
3.	G-3	8.24	80	80

8.2 Noise at operator's ear level

Date of test	: 19.02.2025
Type of track	: Soil surface
Background noise level, dB (A)	: 52
Atmospheric conditions	
Temperature, °C	: 33.5
Pressure, kPa	: 98.8
Relative humidity, %	: 28.4
Wind velocity, m/s	: 1.0 to 1.4
Observed noise level, dB (A)	: 95

9. AIR CLEANER OIL PULL OVER TEST

Date : 20.02.2025

Range of atmospheric condition

Temperature, °C	: 29.6 to 30.9
Relative humidity, %	: 48.2 to 49.4
Pressure, kPa	: 98.6 to 98.8

Mass of oil in the air cleaner assembly : 52.5
when filled with recommended grade of
oil 5% in excess than mark level, g

Sr. No.	Position	Slope (°)	Loss of oil (g)	Oil pull over (%)	Remarks if any
1.	Horizontal	0°	0.1	0.19	None
2.	Tilt longitudinally with front end up	15°	0.2	0.38	
3.	Tilt longitudinally with rear up	15°	0.5	0.95	
4.	Tilt laterally with right side up	15°	0.3	0.57	
5.	Tilt laterally with left side end up	15°	0.2	0.38	



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10. HARDNESS & CHEMICAL COMPOSITION OF BLADES

Hardness & chemical analysis of primary element of tilling and earthing up blades were carried out as per IS: 6690-1981(Reaffirmed 2022). The detail of same is given in table 4 to 7.

10.1 Table 4: Hardness of tilling blade

	Requirement as per IS: 6690-1981 (Reaffirmed 2022) (HRC)	Hardness (HRC) as observed	Remarks
At edge portion	56 ± 3	46 to 48	Does not conform
At shank portion	37 to 45	40 to 43	Conforms

10.2 Table 5: Hardness of earthing up blade

	Requirement as per IS: 6690-1981 (Reaffirmed 2022) (HRC)	Hardness (HRC) as observed	Remarks
At edge portion	56 ± 3	44 to 46	Does not conform
At shank portion	37 to 45	41 to 43	Conforms

10.3 Table 6 : Chemical analysis of tilling blade

Constituents	As per IS: 6690-1981 (Reaffirmed 2022)		As per Ministry's communication No. 13-9/2019 M&T (I&P) dated 26.04.2019	Composition as observed	Remarks
	Carbon Steel	Silicon Manganese steel			
Carbon (C)	0.70 -0.85	0.50-0.60	0.24-0.30	0.91	Does not conform
Silicon (Si)	0.10 -0.40	1.50-2.00	0.40 (Max.)	0.50	Does not conform
Manganese (Mn)	0.50 -1.00	0.50-1.00	1.10-1.40	1.06	Does not conform
Sulphur (S)	0.05 (Max.)	0.05 (Max.)	0.035 (Max.)	0.04	Conforms
Phosphorous (P)	0.05 (Max.)	0.05 (Max.)	0.025 (Max.)	0.01	Conforms

10.4 Table 7 : Chemical analysis of earthing up blade

Constituents	As per IS: 6690-1981 (Reaffirmed 2022)		As per Ministry's communication No. 13-9/2019 M&T (I&P) dated 26.04.2019	Composition as observed	Remarks
	Carbon Steel	Silicon Manganese steel			
Carbon (C)	0.70 -0.85	0.50-0.60	0.24-0.30	0.84	Conforms to carbon steel
Silicon (Si)	0.10 -0.40	1.50-2.00	0.40 (Max.)	1.00	Does not conform
Manganese (Mn)	0.50 -1.00	0.50-1.00	1.10-1.40	0.72	Conforms to carbon steel & silicon manganese steel
Sulphur (S)	0.05 (Max.)	0.05 (Max.)	0.035 (Max.)	0.02	Conforms
Phosphorous (P)	0.05 (Max.)	0.05 (Max.)	0.025 (Max.)	0.01	Conforms

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11. RUNNING – IN

The power weeder was run-in for 0.50 hour before the field performance test as recommended by the applicant. All the fasteners were checked and tightened thereafter.

12. FIELD TEST

The field tests under dry land condition were conducted for 13.43 & 12.85 hours with tilling blade and earthing up device respectively. The field tests were conducted at the recommended field speed of 3600 rpm. In all, 6 test trails i.e. 3 with tilling blade and 3 with earthing up device were conducted in sandy loam soil at N.R.F.M.T.T.I Farm, Hisar. The summary of the field test for dry land operation is given in table-6.

- | | | | |
|-----|--------------------|---|----------------|
| i) | Type of weed | - | Seasonal weeds |
| ii) | Height of weed, cm | - | 8 to 36 |

Table 6: SUMMARY OF FIELD PERFORMANCE TEST

Sr. No.	Parameter		Tilling blade	Earthing up
i)	Type of soil	:	Sandy loam	Sandy loam
ii)	Soil moisture, %	:	10.6 to 11.1	9.7 to 10.1
iii)	Bulk density of soil, g/cc	:	1.21 to 1.32	1.17 to 1.22
iv)	Speed of operation, kmph	:	1.61 to 1.62	1.30 to 1.35
v)	Depth of cut, cm	:	4 to 5	13
vi)	Width of cut, m	:	0.81 to 0.85	1.14 to 1.16
vii)	Area covered, ha/h	:	0.122 to 0.126	0.118 to 0.126
viii)	Time required for one ha, h	:	7.94 to 8.23	7.94 to 8.93
ix)	Fuel consumption			
		l/h :	1.02 to 1.04	0.95 to 1.10
		l/ha :	8.10 to 8.59	8.48 to 9.00
x)	Weeding efficiency, %	:	90.4 to 92.9	--
xi)	Field efficiency, %	:	91.3 to 93.6	73.7 to 81.8

13. ADJUSTMENT, DEFECTS, BREAKDOWNS & REPAIR

No noticeable defect/breakdown was observed during test.

14. COMPONENTS/ASSEMBLY INSPECTION AND ASSESSMENT OF WEAR

14.1 Engine:

The engine and other assemblies were dismantled after 36.78 hours of engine operation.

14.1.1 Cylinder:

Cylinder bore dia. (mm)

Top position		Middle position		Bottom position		Max. permissible wear limit
Thrust	Non-thrust	Thrust	Non-thrust	Thrust	Non-thrust	
86.02	86.00	86.02	86.00	86.02	86.00	



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14.1.2		Piston:				
Piston diameter (mm)						
Top position		At skirt			Max. permissible wear limit (mm)	
Thrust side	Non-thrust side	Thrust side	Non-thrust side	Piston to cylinder clearance (mm)	Piston dia. at skirt	Piston to cylinder clearance
85.77	85.71	85.88	Not measured due to piston design constraint	0.14	85.70	0.48

14.1.3 Piston Rings end gap:

Ring No.	Ring end gap (mm)			Max. permissible wear limit (mm)
	At top	At middle	At bottom	
1 st compression ring	0.20	0.25	0.30	1.00
2 nd compression ring	0.25	0.30	0.30	1.00
Oil ring	0.25	0.30	0.25	1.00

14.1.4 Big end bearing

Dia. of crank pin (mm)	Dia. of bearing (mm)	Clearance (mm)		Max. permissible wear limit (mm)	
		Diametrical	Axial	Diametrical	Axial
40.10	40.06	0.04	0.25	0.045	0.40

14.1.5 Main bearing of crank shaft:

Sr. No.	Dia. of main Journal (mm)	Dia. of main bearing (mm)	Diametrical Clearance of main bearing	End float of crank shaft	Max. permissible wear limit (mm)	
					Diametrical	End float of crank shaft
1.	39.96	39.45	0.51	0.40	0.60	0.75
2.	Ball bearing is provided hence not applicable					

14.1.6 Piston Rings groove clearance:

Ring No.	Ring groove clearance (mm)	Max. permissible wear limit (mm)
1 st compression ring	0.04	0.15
2 nd compression ring	0.04	0.15
Oil ring	0.04	0.15

14.1.7 Valve guide clearance:

Valve guide diameter (mm)		Valve stem diameter (mm)		Valve guide clearance (mm)		Max. permissible wear limit (mm)	
Inlet	Exhaust	Inlet	Exhaust	Inlet	Exhaust	Inlet	Exhaust
7.02	7.07	6.96	6.93	0.08	0.14	0.20	0.30



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14.2	Valves, guides and timing gear	Observation
	Any marked sign of overheating of valves :	None
	Pitting of seat/faces of valves :	None
		Spring stiffness (N/mm) Discard limit (N/mm)
	Inlet valve spring :	11.7 10.0
	Exhaust valve spring :	10.9 10.0
14.3	Timing gears :	No noticeable defect was observed
14.4	Transmission :	No noticeable defect was observed
14.5	Rotary drive unit :	No noticeable defect was observed

14.6 Wear of blades:

14.6.1 Mass basis:

The wear of the tilling blades was measured after 13.93 hrs. of field operation and the observations are as under:

Sr. No.	Initial mass (g)	Mass after 13.93 hrs. (g)	Loss of mass (g)	Percent wear (%)	Percent wear per hour (%)
1.	311.8	306.1	5.7	1.83	0.13
2.	303.0	297.9	5.1	1.68	0.12
3.	317.2	313.5	3.7	1.17	0.08
4.	280.8	275.9	4.9	1.75	0.13
5.	293.1	288.9	4.2	1.43	0.10
6.	304.9	300.0	4.9	1.61	0.12

The wear of the earthing up blades was measured after 12.85 hrs. of field operation and the observations are as under:

Sr. No.	Initial mass (g)	Mass after 12.85 hrs. (g)	Loss of mass (g)	Percent wear (%)	Percent wear per hour (%)
1.	297.0	292.1	4.9	1.65	0.13
2.	305.9	301.8	4.1	1.34	0.10
3.	345.6	342.2	3.4	0.98	0.08
4.	274.0	269.7	4.3	1.57	0.12
5.	313.3	310.7	2.6	0.83	0.06
6.	346.1	342.1	4.0	1.16	0.09



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15. CRITICAL TECHNICAL SPECIFICATIONS

(Vide Ministry's communication F. No. 9-1/2019-M&T (I&P) Dated 20.08.2019)

Sr. No.	Parameters	Specifications	Observed		Remarks
			Tilling blade	Earthing up blade	
1.	Type	Self-propelled, walk behind	Self propelled, walk behind type		Conforms
2.	Working width, mm	300-1500	870	315	Conforms
3.	Type of engine	Compression/Spark ignition	Compression ignition		Conforms
4.	Starting method	Manual/recoil/self-starting	Manual, recoil & self-starting		Conforms
5.	Type of clutch	Dry/Wet/Tensioner pulley	Wet		Conforms
6.	Type of primary gear box	Sliding/constant mesh or combination of both	Combination of both		Conforms
7.	Type of secondary gear box	Gear type, chain & sprocket type	Gear type and chain & sprocket		Conforms
8.	Material for rotor shaft	SAE 1045 (CRS) / EN8 / EN9	SAE 1045		Conforms
9.	No. of flanges	4 - 10	6	--	Conforms
10.	Types of flanges	Square/circular/rectangular	Square	--	Conforms
11.	Distance between consecutive flanges, mm	80 to 150	135	--	Conforms
12.	No. of blades in each flange	3-6	4	--	Conforms
13.	No. of rotor blade	12 (min.)	24	10	Conforms
14.	Thickness of rotor blade, mm	5 (min.)	5.0	6.1	Conforms
15.	Material of blade	Boron (28MnCrB5) / High carbon steel EN 42J	EN 42J		Conforms
16.	Hardness of blade, HRC	38 (min.)	40 to 48	41 to 46	Conforms
17.	Shape of rotor blade	C / J shape	J	C	Conforms
18.	Provision for handle height adjustment	Must be provided	Provided		Conforms
19.	Provision for handle rotation	Must be provided	Provided		Conforms
20.	Provision for emergency stop of engine	Must be provided	Provided		Conforms
21.	Provision for easy start of engine	Must be provided	Provided		Conforms

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22.	Provision for shield/cover to prevent flying of mud & stone from rotor	Must be provided	Provided	Conforms
23.	Depth control mechanism	Must be provided	Provided	Conforms
24.	Provision for transport wheels	Must be provided	Provided	Conforms
25.	Provision for cover on exhaust	Must be provided	Provided	Conforms
26.	Direction of exhaust emission away from operator	Must be provided	Provided	Conforms
27.	Marking/labeling machine	The labeling plate should be riveted on the body of machine having Name and address of manufacturer & Applicant, Country of origin, Make, Model, Year of manufacturer, Serial number, Engine number, Engine HP, rated rpm & SFC.	Provided, except country of origin	Partially conform
28.	Literature	Operator manual, service manual and Parts catalogue should be provided.	Provided	Conforms

16. COMMENTS & RECOMMENDATIONS

16.1 Engine Performance

- The maximum power & rated power of engine was observed as 7.08 & 6.84 kW against the declaration of 6.40 kW for both maximum and rated power.
- The specific fuel consumption corresponding to max. power is observed as 485.88 g/kWh against the declaration of 280 g/kWh which is on higher side and should be looked into for corrective action.

16.2 Mechanical vibration

The amplitude of mechanical vibration marked as (*) in the relevant chapter, are on higher side. It is not just directly concerned with operator's health, safety and comfort, but also adversely affects the useful life of the components. In view of above, this deserved to be given top priority for necessary corrective action.



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16.3 The chemical composition of tilling and earthing up blades does not conform to the requirements of IS: 6690-1981 (Reaffirmed 2022). This needs to be looked into for necessary corrective action.

16.4 The hardness of tilling and earthing up blades does not conform to the requirements of IS: 6690-1981 (Reaffirmed 2022). This needs to be looked into for necessary corrective action.

16.5 Field performance test

No noticeable defect was observed during field test.

16.5.1 Tilling blade

A. Rate of work

- i. The speed of operation ranged from 1.61 to 1.62 kmph during the test. The rate of work varied from 0.122 to 0.126 ha/h.
- ii. The fuel consumption varied from 1.02 to 1.04 l/h.
- iii. The fuel consumption per unit area of operation varied from 8.10 to 8.59 l/ha.

B. Quality of work

- i. Weeding efficiency of machine ranges from 90.4 to 92.9 % and the field efficiency ranges from 91.3 to 93.6 %.

16.5.2 Earthing up blade

A. Rate of work

- iv. The speed of operation ranged from 1.30 to 1.35 kmph during the test. The rate of work varied from 0.118 to 0.126 ha/h.
- v. The fuel consumption varied from 0.95 to 1.10 l/h.
- vi. The fuel consumption per unit area of operation varied from 8.48 to 9.00 l/ha.

B. Quality of work

- ii. The field efficiency ranges from 73.7 to 81.8 %.

16.6 Labeling plate

The labeling plate riveted on machine partially meets the requirement of Critical Technical Specification. The country of origin shall also be included in the labeling plate.



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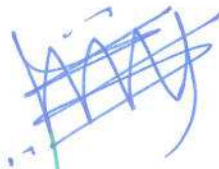
17. TECHNICAL LITERATURE

The following literatures were provided by the applicant.

- User manual
- Parts catalogue
- Service manual

However, operator's manual needs to be updated as per IS: 8132-2023.

TESTING AUTHORITY

Er. C. K. TIJARE AGRICULTURAL ENGINEER	
DR. MUKESH JAIN DIRECTOR	 07.04.2025

The draft test report has been prepared by Sh. Vikram, Senior Technician

18. APPLICANT'S COMMENTS

Para No.	Our reference	Applicant comments
18.1	16.1 (ii)	We will work on comments for better result of fuel consumption, In support with our R & D and production team. We will make all possible necessary changes in our product future.
18.2	16.2	We will work on it and take necessary steps, in support with our R & D and production team. We will provide anti vibration rubber grips at handle bars for operator comfort, safety and health. We will also look into adding vibration damping gaskets at all joints wherever possible. We make all possible necessary changes in our product in future.
18.3	16.3	We will provide as per the IS. We will make all possible necessary changes in our product in future.
18.4	16.4	We will provide as per the IS. We will make all possible necessary changes in our product in future.



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ANNEXURE – I

FIELD DATA SHEET OF TILLING BLADE

Place: NRFMTTI Farm, Hisar
Type of soil: Sandy loam

Test No.	Date	Duration of test (h)	Speed of operation (kmph)	Avg. width of cut (m)	Avg. depth of cut (cm)	Avg. soil moisture (%)	Bulk density (g/cc)	Fuel consumption		Area covered ha/h	Time required for one hectare h/ha	Field efficiency (%)	Weeding efficiency (%)
								l/h	l/ha				
1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.	11.02.2025	2.50	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
2.	12.02.2025	3.39	1.62	0.85	4	10.9	1.29	1.02	8.10	0.126	7.94	91.3	92.9
3.	12.02.2025	3.41	1.61	0.83	5	11.1	1.21	1.02	8.23	0.123	8.11	92.5	91.3
4.	13.02.2025	4.13	1.61	0.81	5	10.6	1.32	1.04	8.59	0.122	8.23	93.6	90.4

FIELD DATA SHEET OF EARTHING UP BLADES

1.	13.02.2025	1.58	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
2.	14.02.2025	3.40	1.32	1.16	13	9.7	1.17	0.95	8.48	0.120	8.93	73.68	NA
3.	14.02.2025	3.50	1.30	1.15	13	9.9	1.20	1.06	9.00	0.118	8.47	78.67	NA
4.	18.02.2025	4.37	1.35	1.14	13	10.1	1.22	1.10	8.73	0.126	7.94	81.82	NA

